Let’s Make Music with Programming!
An Introduction to TidalCycles (for Beginners) +
An Audio–Visual Live Performance System

SESSIONS 2024

moistpeace 2024/11/17

Introduction

- Today, I’ll introduce the first step toward making music with programming.
- If you think, “This sounds fun” or “I want to try it,” we’ve succeeded.
- I’m not a programmer, so please forgive any gaps in technical depth.
- In the second half, I’ll show how to synchronize the audio and visuals used in yesterday’s live system.

Today’s Schedule

- 10:30–10:35 Introduction / About Me
- 10:35–10:55 Part I: An introduction to tools for making music with programming
- 10:55–11:15 Part II: An introduction to a live system that synchronizes audio and visuals
- 11:15–11:20 Questions & Answers

About Me

moistpeace

- Creates beats and visuals through programming, and performs audio/visual live sets
- Uses TidalCycles, Ableton Live, TouchDesigner, and openFrameworks
- Explores systems that synchronize audio and visuals
- Currently exploring the use of AI
- Born in Nagano; lived in Fukuoka and Hyogo; now based in Tokyo
- Electrical engineer at a manufacturer; circuit engineer for headphones
- Artists I would love to open for: Flying Lotus, Squarepusher, Aphex Twin, Max Cooper, and Ryoji Ikeda





Appearances

[Live]

TouchDesigner meetup TOKYO 2022.11

Hyper geek #4 SUPER DOMMUNE

Hyper geek #6

nonlinear-nauts [exp.020, 022]

Audio Visual Event “draw(tokyo);”

n/a (at VRChat)

TOKYO XR, Metaverse & Content Business World

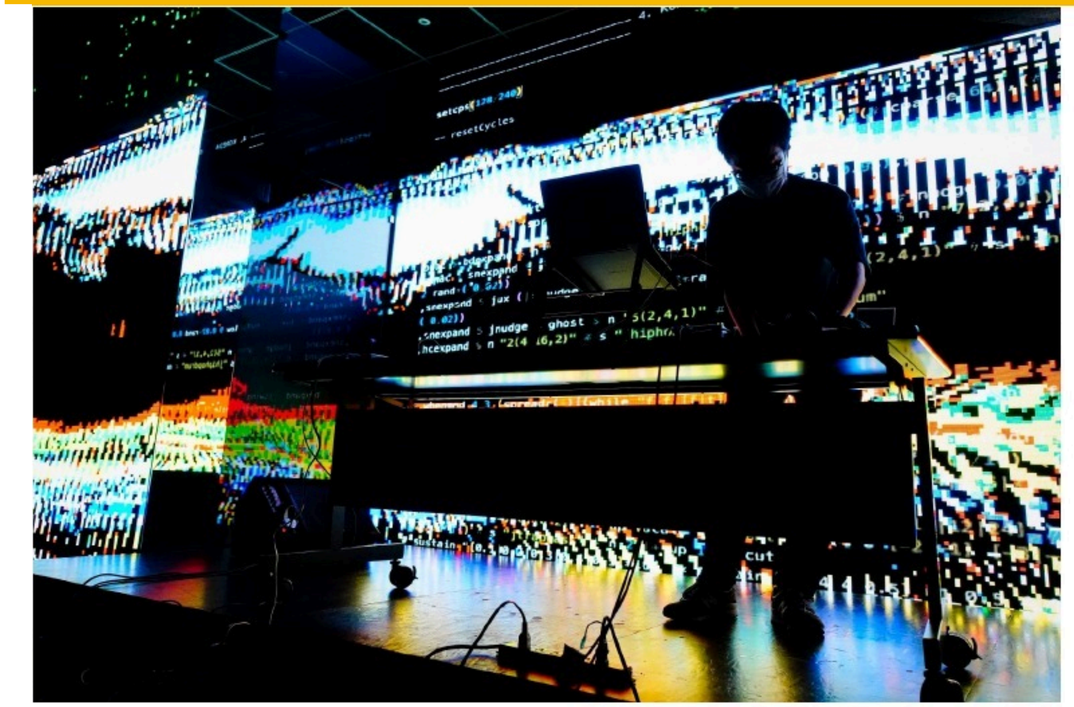
TDF 16ms #0

GYOEN GYOEN GYOEN

YCAM sound tectonics #27 “Coding Discussion”

AMCJ “Deep Live Coding”

Booking inquiries welcome



[Contest]

Beat Grand Prix 2020: 3rd / 2021: 2nd / 2022: 6th / KM BEAT CYPHER Finalist

Promotion

AMCJ “Deep Live Coding” — streaming tickets now on sale

I taught a 2.5-hour intermediate TidalCycles class, plus a mini live set



Archive stream available until 11:00 p.m. on Sunday, November 24

Part I

Making Music with Programming

Within programmatic music production,

I'll focus on live coding.

What Is Live Coding?

- A performance in which music and visuals are improvised in real time through programming

- The computer screen is usually projected onto a display

- Algorave: a coined term combining “algorithm” and “rave”

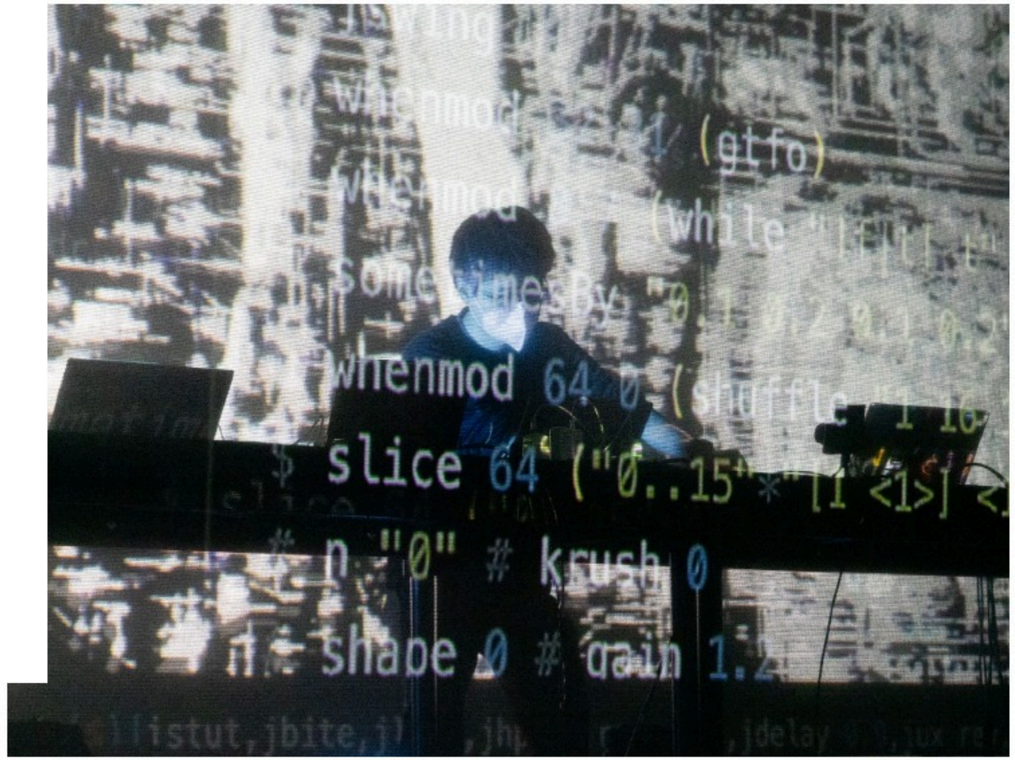


Photo: Atsushi Tanabe

Photo courtesy of Yamaguchi Center for Art and Media [YCAM]

Tool Overview

- TidalCycles
- Strudel
- Sonic Pi
- FoxDot
- SuperCollider

There are many more options, of course

TidalCycles

- Live-coding environment designed for improvisational performance and composition
- Haskell-based
- Uses SuperCollider

My take
Large live-coding community
Installation can be challenging (though it has improved recently)
Short, readable code



Image source: https://yoppa.org/blog/9561.html

Strudel



- Live-coding environment for music
- Developed as a JavaScript port of TidalCycles
- Runs in the browser, so no installation is required

My take
Recommended if you just want to try something
Also useful if you struggled to set up TidalCycles
Active development; likely to gain more users

```
// "coastline" @by eddyflux
// @version 1.0
samples('github:eddyflux/crate')
setcps(.75)
let chords = chord("<Bbm9 Fm9>/4").dict('ireal')
stack(
  stack( // DRUMS
    s("bd").struct("<[x*<1 2> [~@3 x]] x>"),
    s("~ [rim, sd:<2 3>]").room("<0 .2>"),
    n("[0 <1 3>]*<2!3 4>").s("hh"),
    s("rd:<1!3 2>*2").mask("<0 0 1 1>/16").gain(.5)
  ).bank('crate')
  .mask("<[0 1] 1 1 1>/16".early(.5))
```

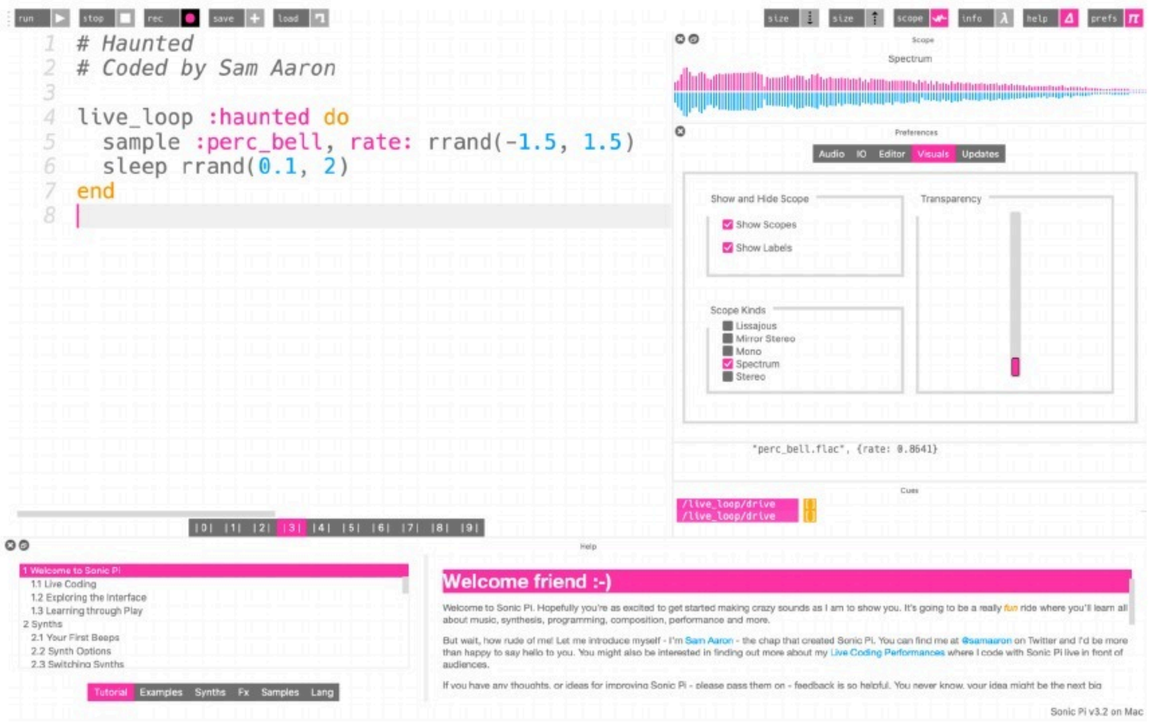
Source: Strudel

https://strudel.tidalcycles.org/workshop/getting-started

Sonic Pi

- Open-source programming environment for sound and music
- Ruby-based
- Extensive documentation and tutorials
- Easy installation

ライミング環境



My take

Recommended if you just want to try something

Large live-coding community

Clear and educational

Source: Wikipedia

https://en.wikipedia.org/wiki/Sonic_Pi

FoxDot

- Live-coding environment using Python
- Uses SuperCollider

My take

Recommended for people comfortable with Python

Might be possible to use other Python libraries to generate vocals?



Source: https://www.youtube.com/watch?v=CXrkq7u69vU

SuperCollider



- A programming environment and language for audio synthesis
- TidalCycles and FoxDot require SuperCollider

My take
Generates the sound itself
Best for people who want to start from sine waves
Not often used on its own for live coding

```
{
  var snare, bdrum, hihat;
  var tempo = 4;

  tempo = Impulse.ar(tempo); // for a drunk drummer replace Impulse with Dust
  snare = WhiteNoise.ar(Decay2.ar(PulseDivider.ar(tempo, 4, 2), 0.005, 0.5));
  bdrum = SinOsc.ar(Line.ar(120,60, 1), 0, Decay2.ar(PulseDivider.ar(tempo, 4
  hihat = HPF.ar(WhiteNoise.ar(1), 10000) * Decay2.ar(tempo, 0.005, 0.5);

  Out.ar(0, (snare + bdrum + hihat) * 0.4 ! 2)
}.play
```

Source: https://supercollider.github.io/

I introduced many options…

but I keep coming back to TidalCycles.

Why I Love TidalCycles

- Live coding
  - Short, readable code
  - Usable without being a programmer

- Instrument and composition tools
  - Sampler (playback)
  - Sequencer
  - Synthesizer (SuperCollider)
  - MIDI send/receive, synchronization, and CV

- Audio-visual performance
  - OSC (Open Sound Control) output

“Express yourself freely
with short, readable code.”

TidalCycles

I’ll explain it with a live demonstration.

Part I Summary

Which tool caught your interest?

TidalCycles / Strudel / Sonic Pi / FoxDot / SuperCollider

If one catches your eye, just try it out!

Instead of wondering where to start, try them all!

Even if one doesn’t fit you, the time won’t be wasted.

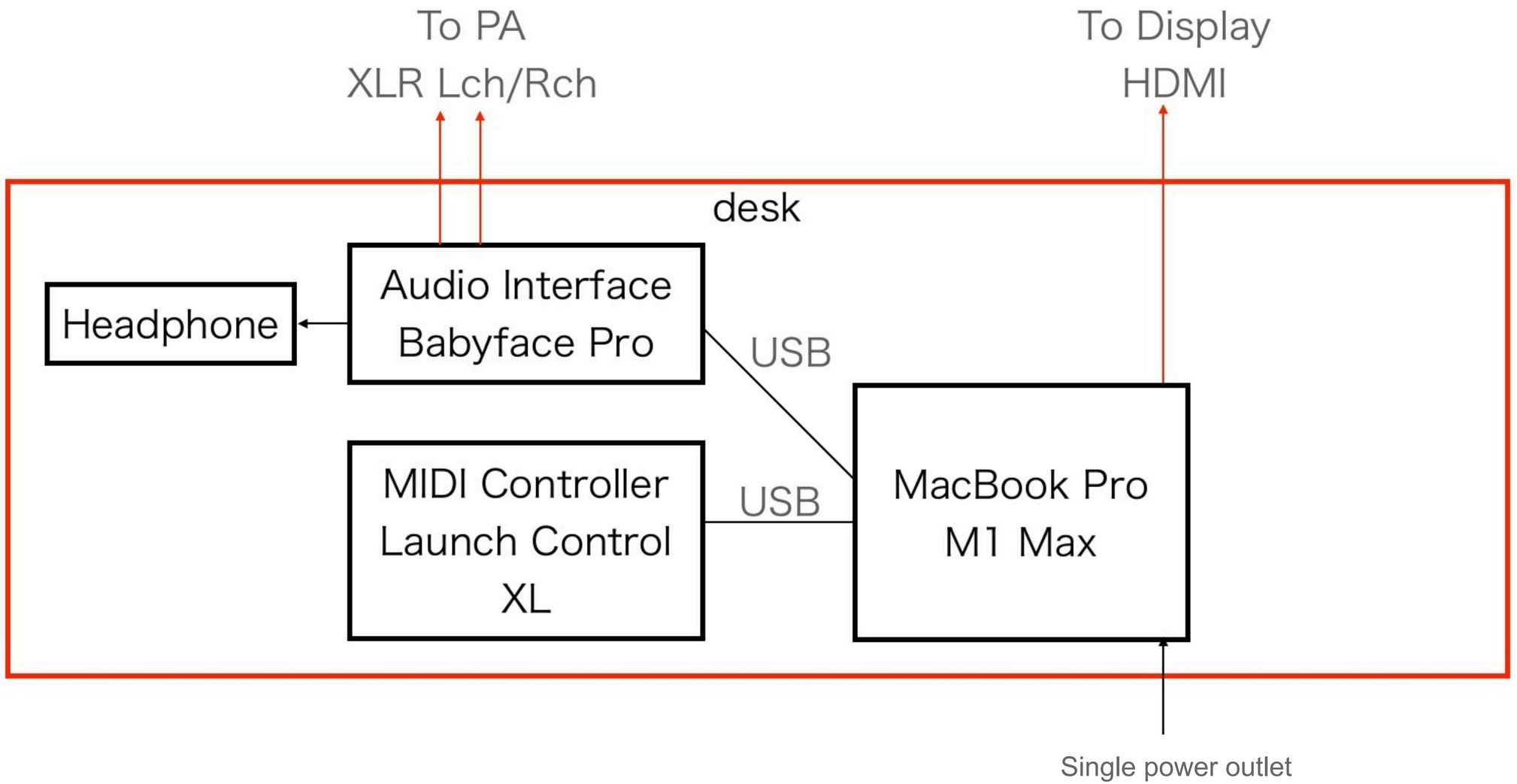
Part II

An Audio–Visual Synchronization System for Live Performance

Live Performance Footage



System Overview: Hardware



System Overview: Software

- Code on screen: TidalCycles (editor: VS Code)
- Music: TidalCycles, Ableton Live
- Visuals: TouchDesigner, openFrameworks
- Visual assets: TD, oF, DALL·E 3, Deforum, Pika, and free stock footage

Music and Visual Synchronization System

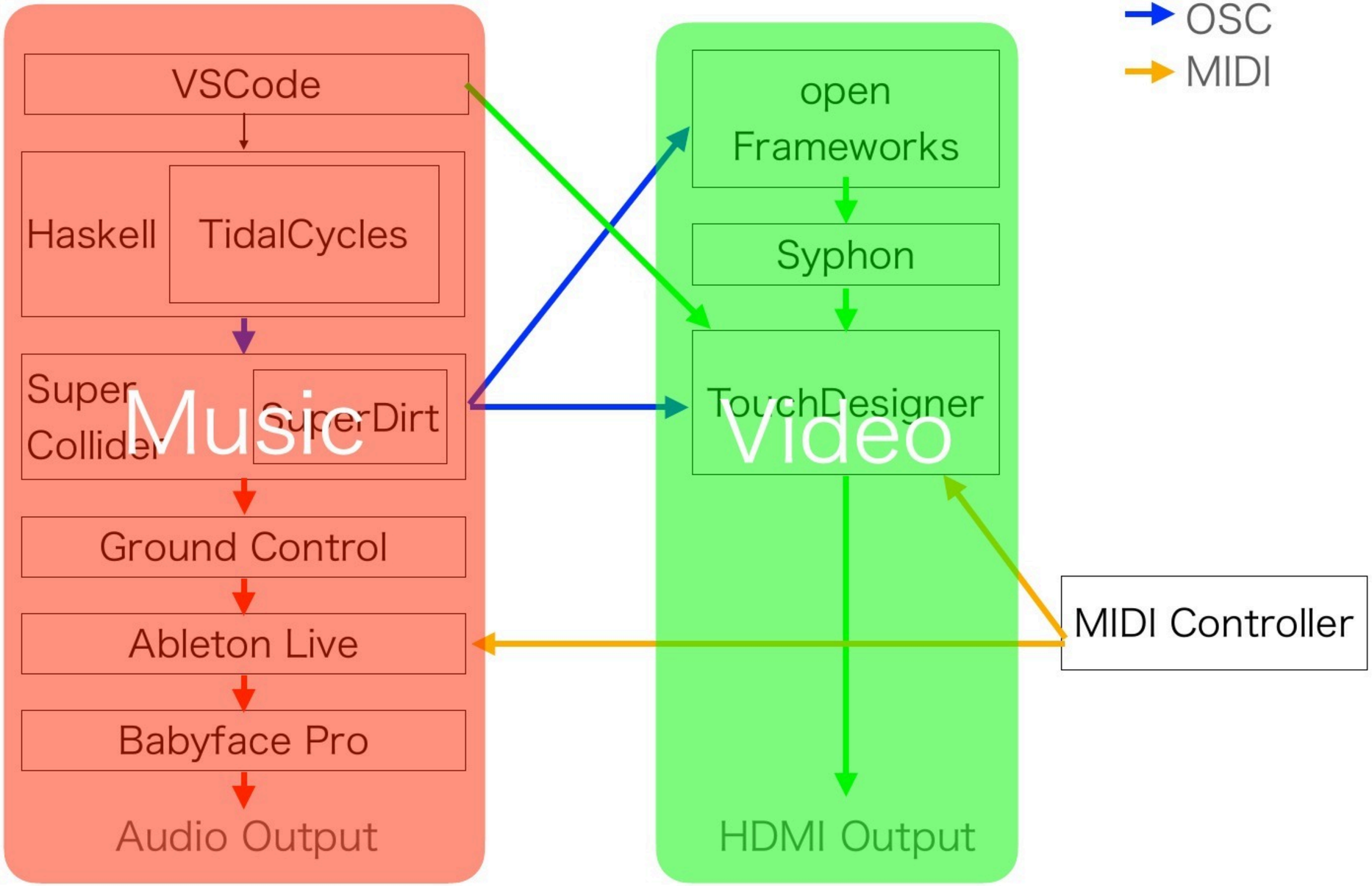
At its core, TidalCycles creates patterns, outputs OSC, and produces sound according to those OSC messages.

→ Feed the OSC into visual software to create visuals matched to the patterns.

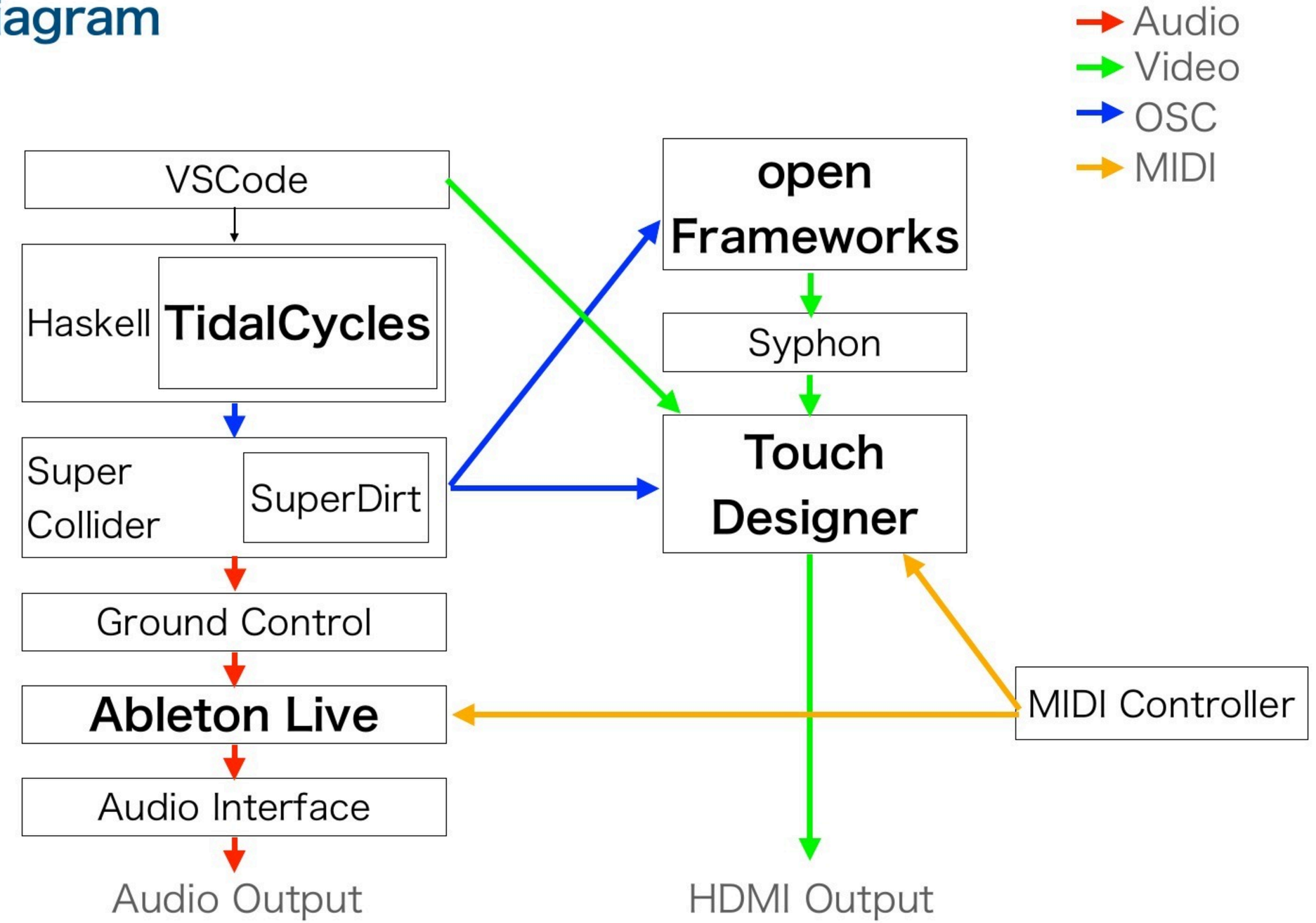
→ The music and visuals synchronize.

OSC: Open Sound Control — a communication protocol

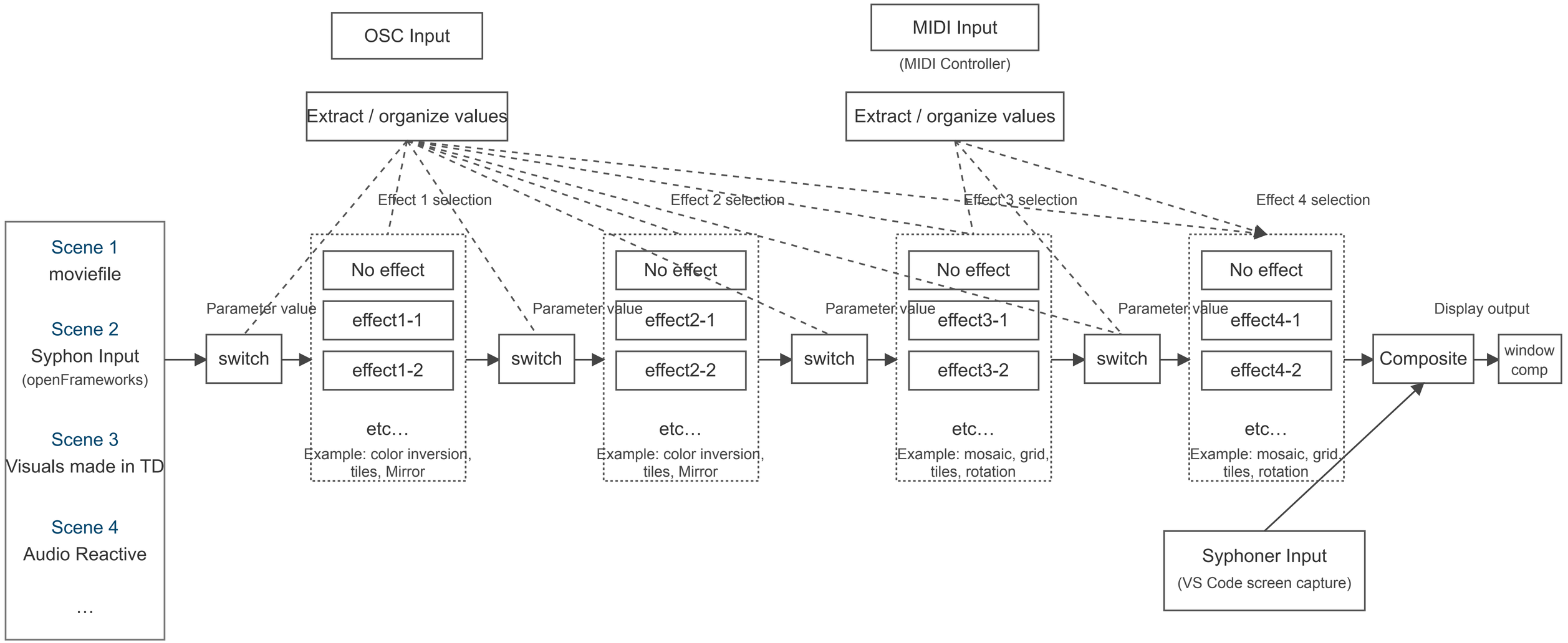
system block diagram



system block diagram



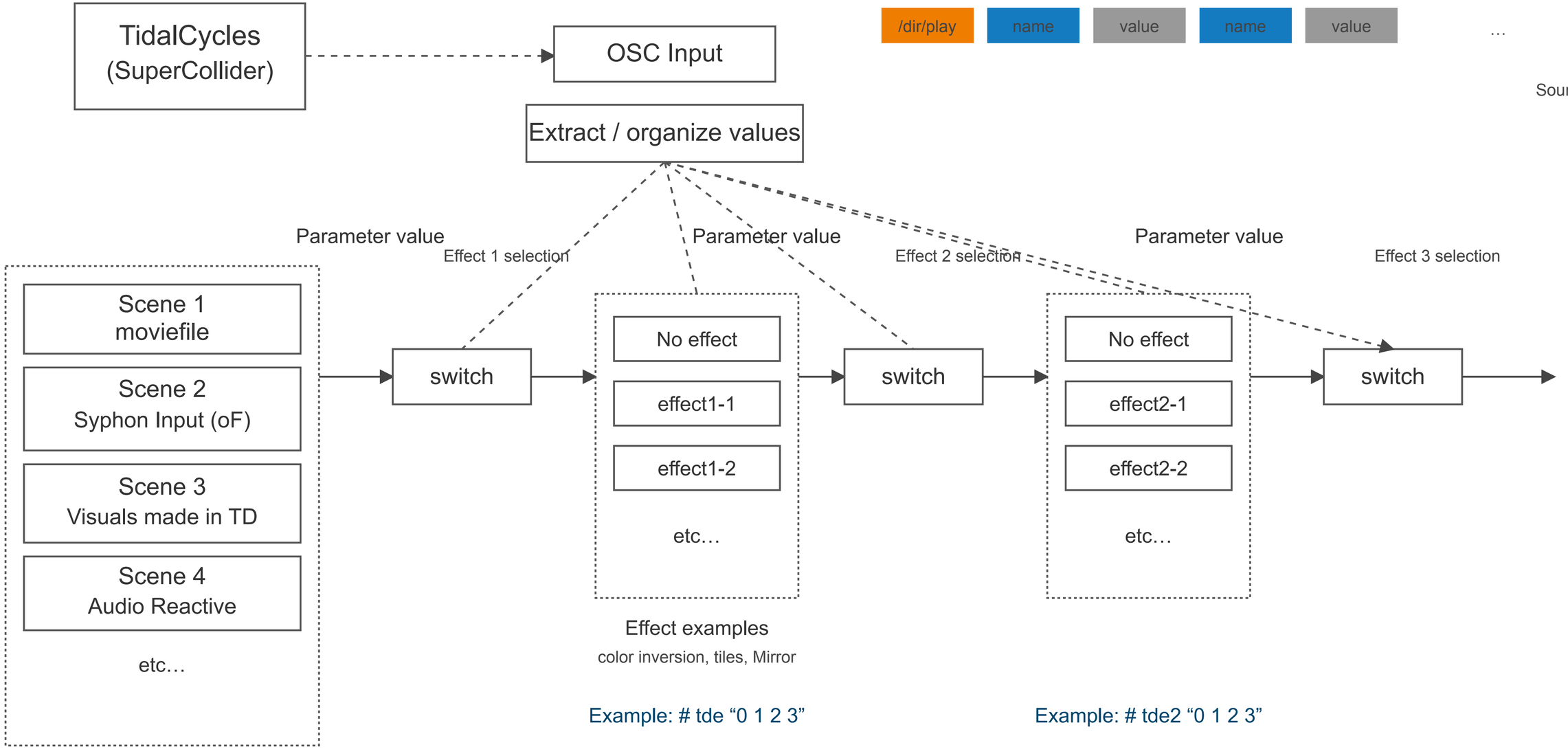
TouchDesigner System Block Diagram



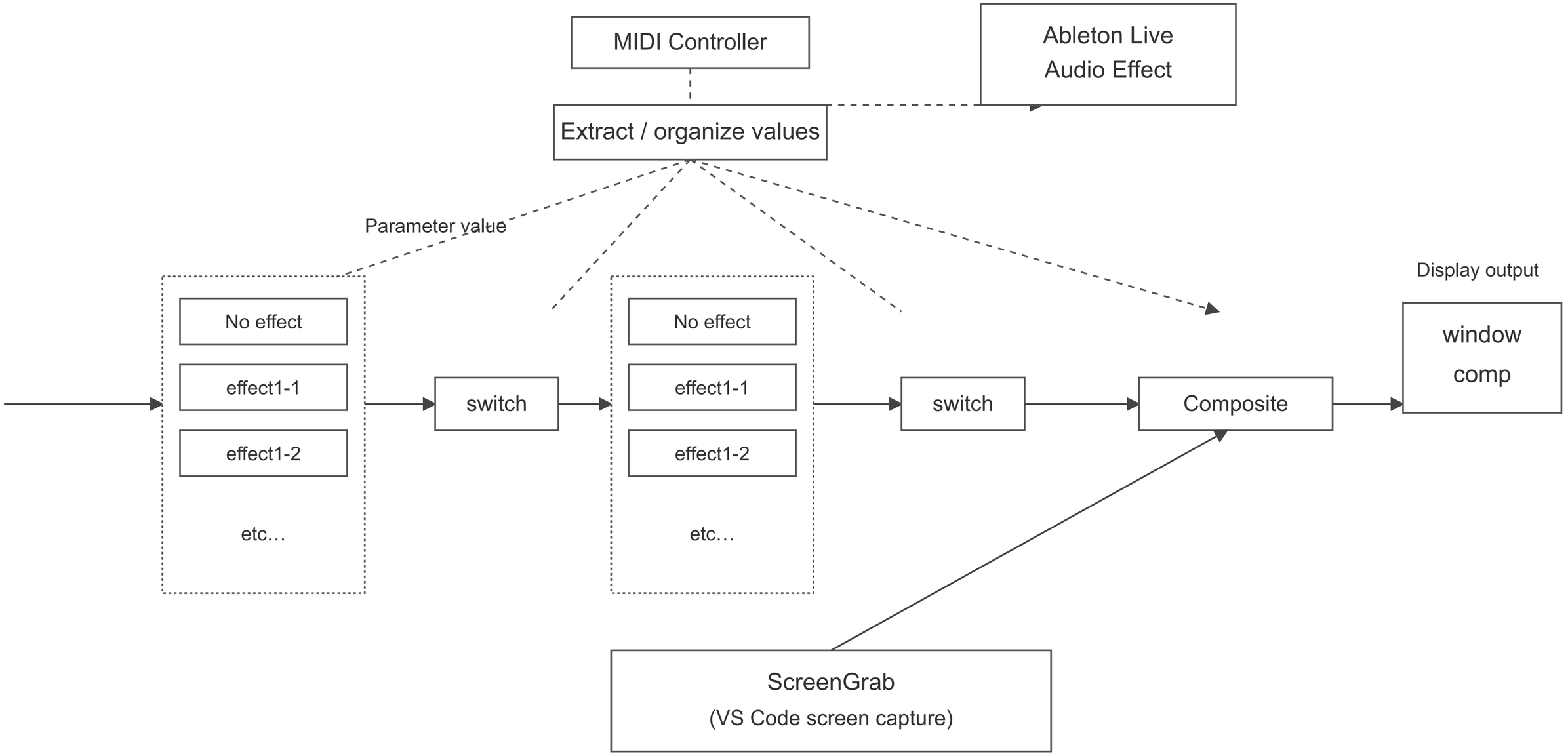
Scenes and effects are placed in Containers and configured to stop cooking when unused, reducing CPU/GPU load. TidalCycles OSC values are extracted and used to switch scenes and effects.

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TouchDesigner System Block Diagram 1 / 2



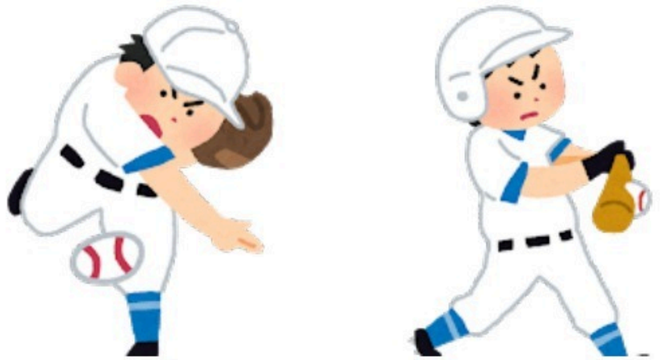
TouchDesigner System Block Diagram 2 / 2



Bonus: Audio–Visual Combinations to Try

る組み合わせ例

- Ableton Live + TouchDesigner
- Ableton Live + Max for Live plugins
- Ableton Live + p5.js
- Max + Jitter
- Modular synthesizer + TouchDesigner
- Groovebox + TouchDesigner
- TidalCycles + TouchDesigner





Thank you for listening!

Show moistpeace some love

and invite me to perform!